FOSS4G HIROSHIMA 2026

Japan Geospatial Times

Exploring the History and Use Cases of Geospatial Technologies in Japan



Eita Horishita, FOSS4G Hiroshima 2026 General Sessions

Self Introduction



Eita Horishita

- FOSS4G 2026 Local Organizing Committee member
- Technology Trend Analyst, Urban Planner & GIS Practitioner
- Passionate about building practical web maps using Open Data

Recent Web Map Projects

- Childcare & Nursery Map
- Urban Planning Map
- Community Blog Map
- Traffic Accident Map

etc.



About Me

https://eitahorishita.com/

Ice Break: Map of Traffic Accidents in Japan



Try the Map!

https://kashiwa.co-place.com/tamap-en/#16/34.3925/132.4531

Large-scale open dataset

Displays approximately **2 million** traffic accident records in Japan from 2019 to 2024.

Trusted open data

Data is published by the **National Police Agency of Japan.**

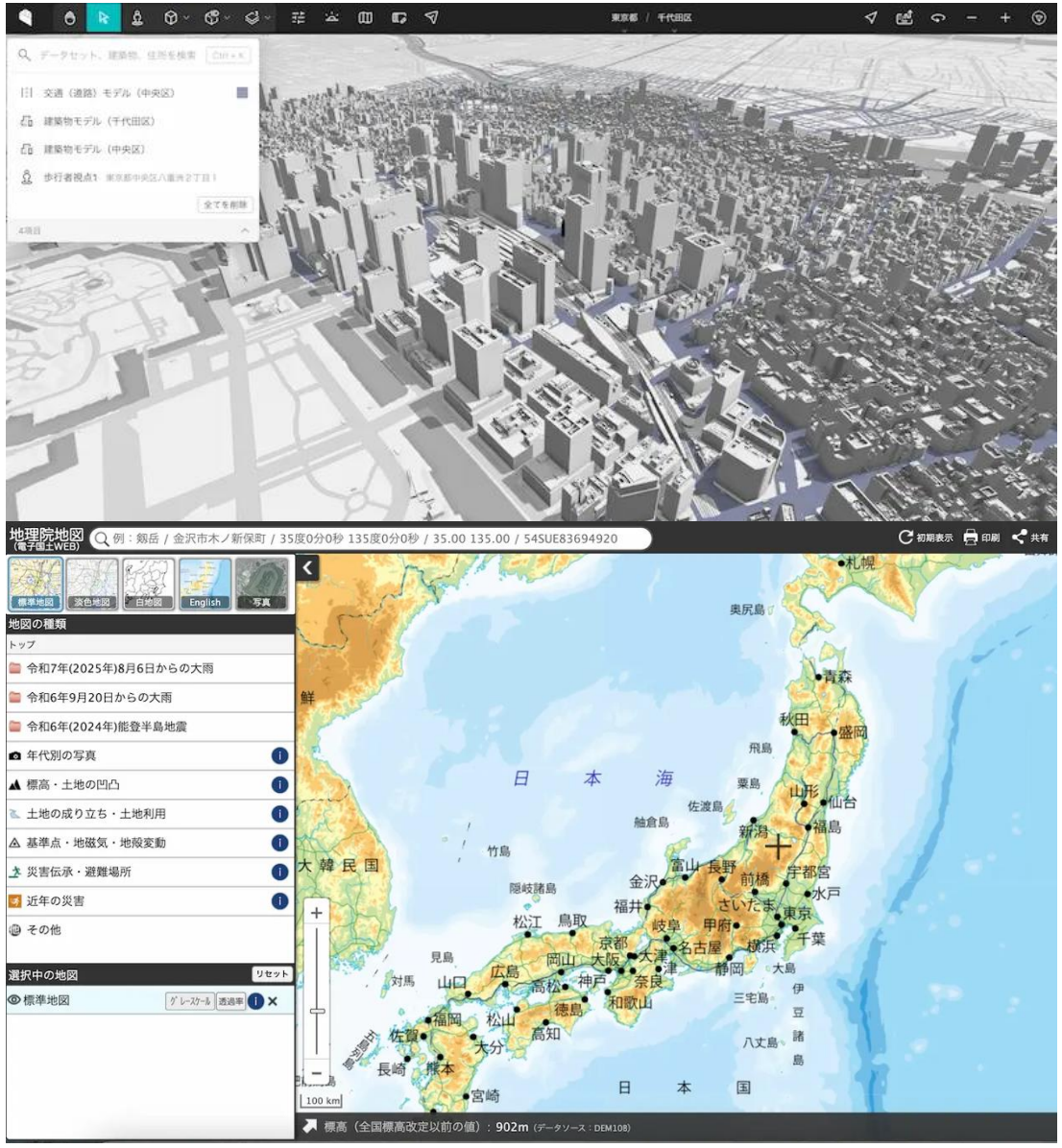
Pinpoint details via pop-ups

Check when, where, and what kind of accidents occurred.

Identify high-risk areas

Helps you spot dangerous locations with a high frequency of traffic accidents.

Is Japan's geospatial ecosystem really so good?



Yes, we have..

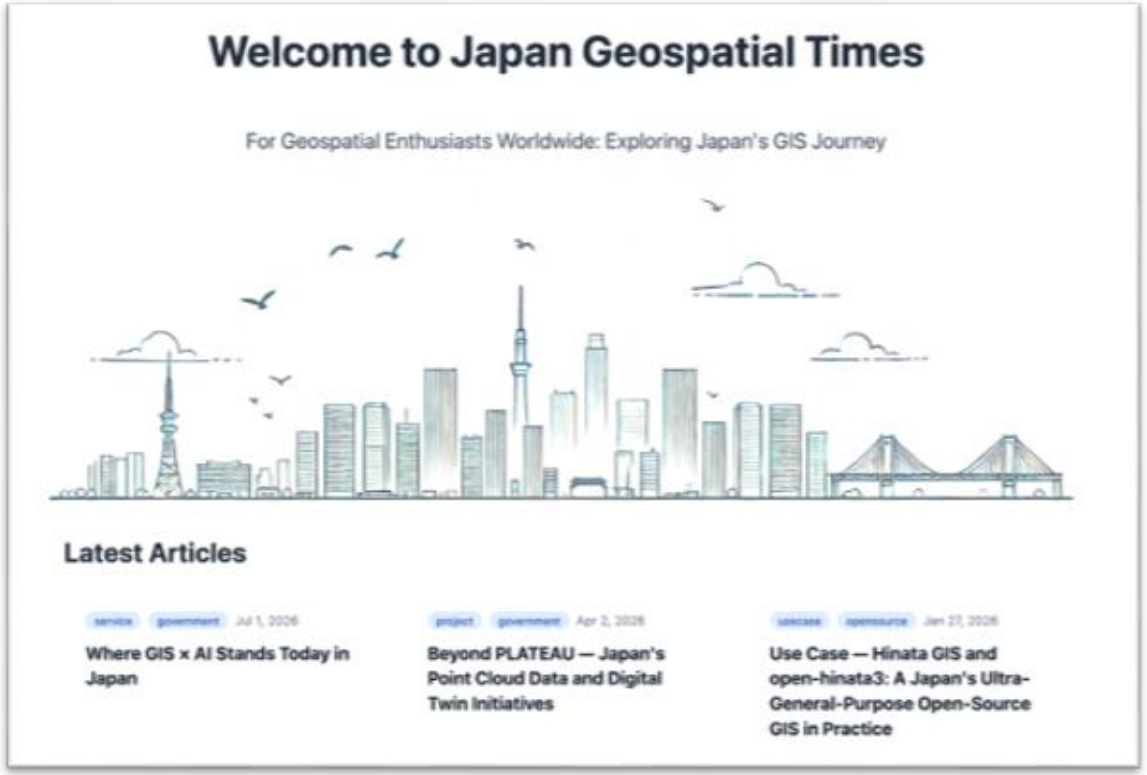
- **Rich open geospatial data**
  e.g. National Land Numeric Information Download Service
- **National platforms for data sharing**
  e.g. G-Spatial Information Center
- **Official web viewers for easy access**
  e.g. GSI Maps
- **National initiatives driving innovation**
  e.g. Project PLATEAU
- **Active open-source communities**
  e.g. OSGeo.JP Code for Japan

But, we have been facing many challenges..

- **Sustainability and high cost of maintaining data and systems**
- **Shortage of skilled professionals**
- **Limited awareness beyond the geospatial community**
- **Keeping up with rapidly evolving technologies**
  e.g. GeoAI, automation, real-time data

Introducing Japan Geospatial Times

To understand where we are and where we've been: Launching **Japan Geospatial Times**.



Visit the site!

https://jgtimes.org/

Explore Japan's geospatial story

A blog platform that shares Japan's geospatial **history, technologies,** and real-world **use cases** for a global audience in English (and other languages).

Built with the spirit of FOSS4G

An **open-source** project that embraces openness, transparency, and community collaboration.

Easy to contribute

Articles written in **Markdown** are automatically converted to HTML, making it simple for anyone to contribute.

Curated resources for deeper understanding

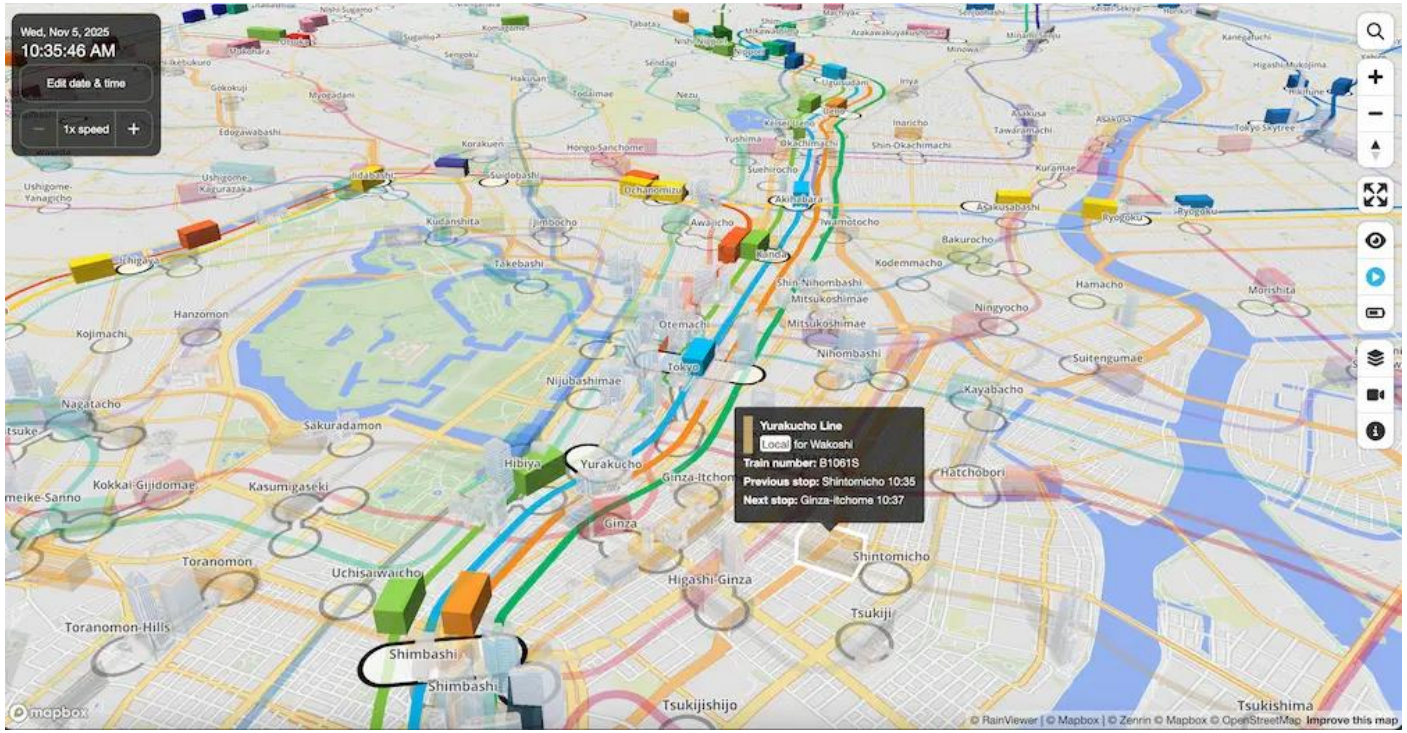
Includes a collection of useful links (**Resources**), author **profiles**, and a Japanese-English **glossary** of technical terms.

Key Milestones in Japan's Geospatial History

1955 – 1973	High Economic Growth Era	**Infrastructure Development** • Rapid economic growth • Massive infrastructure construction (roads, railways, bridges, dams, airports)
1990s	Shift to Maintenance and Management	**Infrastructure Management** • Aging infrastructure becomes a major issue • Limited budgets and resources • GIS introduced for efficient asset management
Turning Point #1 1995	Great Hanshin-Awaji Earthquake	**Disaster Exposes the Need** • Over 6,000 lives lost, massive urban damage • Recognition of the need for shared geospatial information
2000s	Building the Geospatial Infrastructure	**National Spatial Data Infrastructure** • Standardization & legal frameworks • Nationwide geospatial data sharing • Government-led GIS advancement
Turning Point #2 2011	Great East Japan Earthquake	**Disaster Drives Open Data** • Volunteers create maps (OSM, etc.) • Open data policy accelerates
2010s	Open Source & Civic Tech Era	**Open Source & Civic Tech Flourish** • OSGeo Japan promotes open-source GIS • Code for Japan drives civic tech projects • OpenStreetMap community grows • Government × Citizens × OSS collaboration
2020s	AI & Digital Twin Era	**GeoAI & Digital Twins** • AI, big data, and cloud technologies • Digital twins (e.g., PLATEAU) • Smarter cities and societies

Use cases of Open Geospatial Projects in Japan #1

- **Mini Tokyo 3D: A Digital Twin of Tokyo's Public Transportation**



▼ **Live site**

Visit the site!

https://minitokyo3d.com/

Developer

Akihiko Kusanagi

What is Mini Tokyo 3D?

A digital twin application that visualizes the real-time operational status of public transportation around Tokyo.

Background

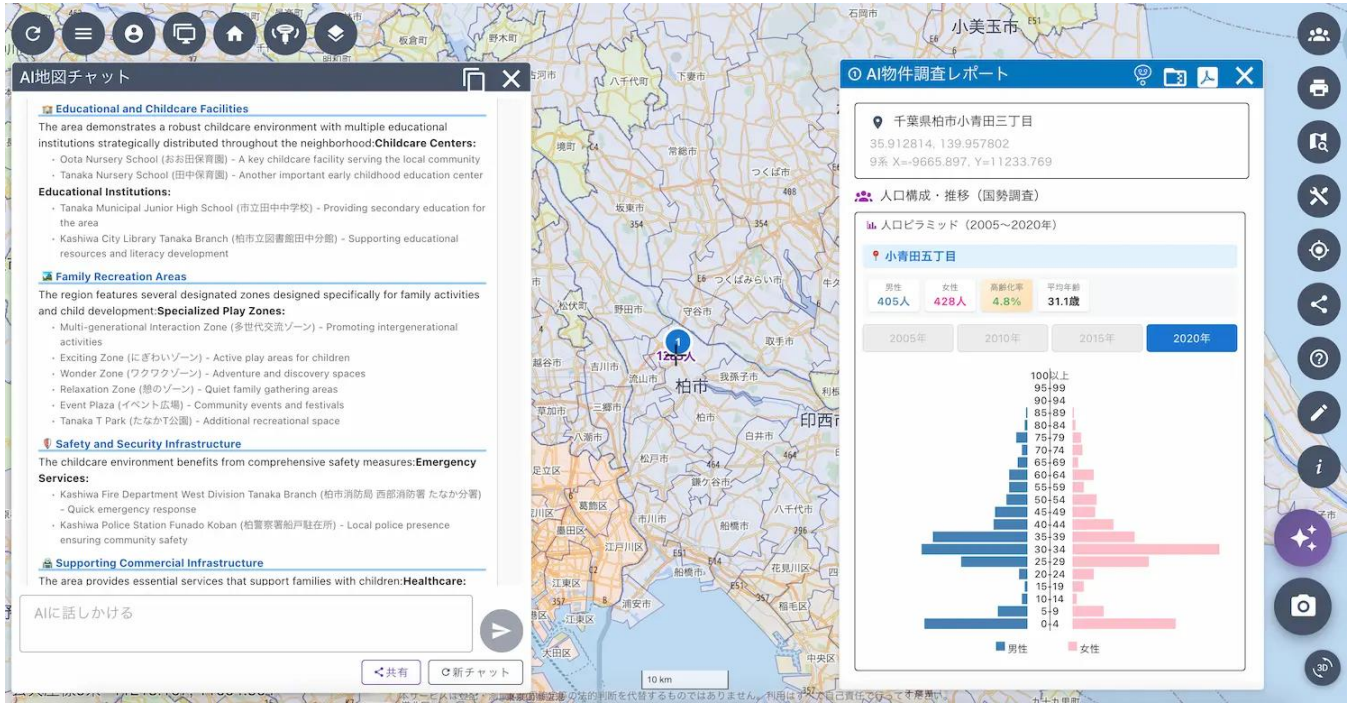
Started from an open data competition in 2019.

Key Features

- Smooth animation of over 2,200 trains operating during rush hours
- Multi-language support (4 languages)
- Built with open data & open-source technologies

Use cases of Open Geospatial Projects in Japan #2

- **Hinata GIS & Open-Hinata3: An Ultra-General-Purpose Open GIS**



▼ **Live site**

Visit the site!

https://kenzkenz.jp/oh3/

Developer

Kenji Ochiai (@kenzkenz)

About Hinata GIS / Open-hinata3

An Ultra-Purpose web GIS

Developed by a Japanese prefectural government employee as a personal project.

Award-winning Project

Won the grand prize at the “RESAS App Contest” (hosted by Japan’s Cabinet Office) in 2017.

Key Strength

Over 600 types of layers with AI agent function

Enables users to integrate and analyze massive layer information from past to present for any location in Japan — in a cross-sectional and AI-assisted manner.

Use cases of Open Geospatial Projects in Japan #3

- **Rekichizu: Rediscovering Japan's History through Modern Web Maps**



▼ **Live site**

Visit the site!

https://rekichizu.jp/en

Developers
Hajime Kato (Graphic Designer) &
Sorami Shiromizu (Software Engineer)

What is Rekichizu?
Rekichizu literally means "*History Map.*"

Concept
Historical maps with modern UX.
Compare past and present in both 2D and 3D.

Main Feature
An interactive web map that recreates Japan during the late Edo period (1600s–1800s) using a modern map design.

Collaboration
Working with digital humanities researchers and publishers to bring history to life.

Summary and Key takeaways

Japan's geospatial data ecosystem has developed primarily for **infrastructure management and disaster response**.

Although there are many datasets, portals, and legal frameworks, the greatest challenge is **continuously updating high-quality, reliable data**.

Since the 2010s, **community-level activities have flourished**, driving many unique and innovative projects.

What's Your Country's Geospatial Times?



Japan Geospatial Times – Technology Stack

Main Component	Role	Benefits	OSS
SvelteKit	Website framework	Lightweight and simple web framework; easy-to-understand file structure	○
Tailwind CSS	Website styling	Clean UI design; responsive and dark-mode support	○
Markdown-it	Article conversion	Automatically converts articles written in Markdown into clean and simple HTML	○
GitHub	Code management	Backup and collaboration	—
Cloudflare	Hosting	Automatic deployment via GitHub; fast delivery worldwide	—